Atom Physics

Lecture 3

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

Lecture	Date	Topics	Material
1	03-09-2025	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	10-09-2025	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	**17-09-2025**	**•Behaviour of a wave at a boundary** **•Confining a particle** **•The Schrödinger equation** **•Applications of the Schrödinger equation**	**Book: Ch. 5.1-5.4**
4	24-09-2025	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

Lecture	Date	Topics	Material
5	01-10-2025	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	08-10-2025	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	15-10-2025	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

Recap Previous Lecture

- Compton Scattering $\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$

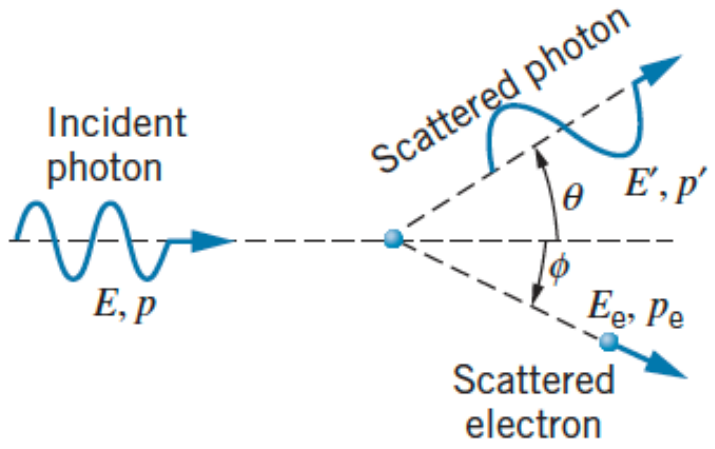


FIGURE 3.18 The geometry of Compton scattering.

- De Broglie Wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$

- Heisenberg's Uncertainty Principles $\Delta x \Delta p_x \geq \hbar$ $\Delta E \Delta t \geq \hbar$

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Today:

Schrödinger Equation

$$i\hbar\frac{\partial\psi(\mathbf{r},t)}{\partial t} = \hat{H}\,\psi(\mathbf{r},t)$$

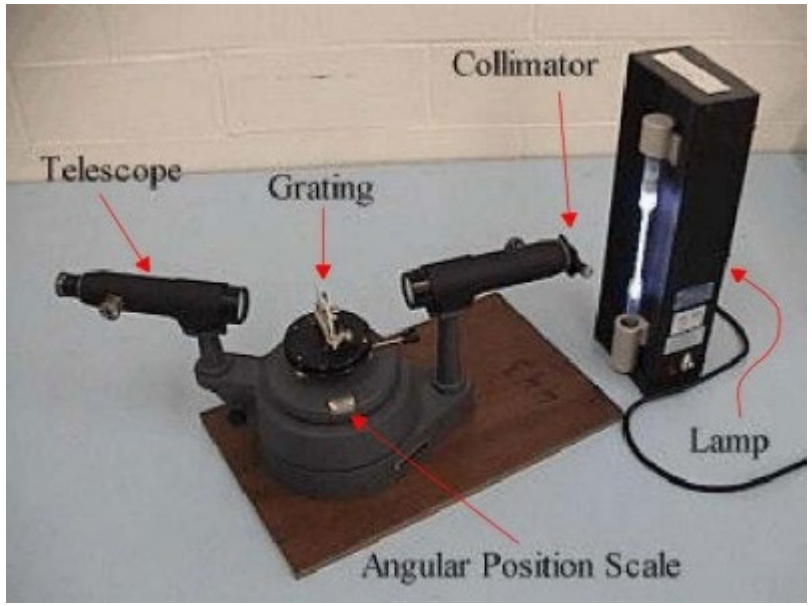
Time-dependent Schrödinger equation

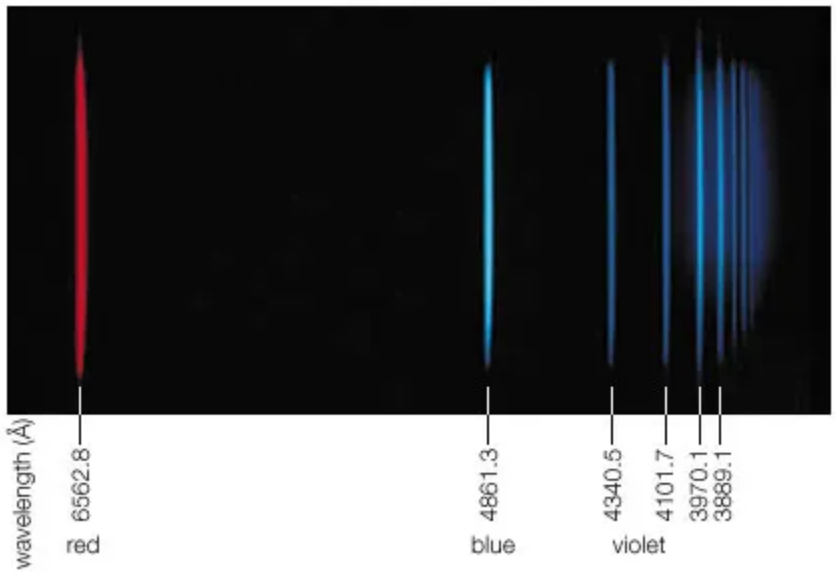
This equation tells how quantum states evolve in time, not what will happen - only what can happen.



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- Schrodinger published his equation in 1925 in an attempt to provide a framework that explains energy transitions as observed in spectroscopy





(Note: we take a jump in time here: Bohr had already postulated his 'orbit model' in 1913- we'll discuss that next week)

Describing wave packets in time and space

Classical *particle*: $F = -\frac{dU(x)}{dx} = m\frac{d^2r(x)}{dt^2}$ (r: particle position)

Classical *wave*: $\frac{\partial^2 u(x,t)}{\partial t^2} = v^2\frac{\partial^2 u(x,t)}{\partial x^2}$ (u: wave amplitude)

Can we come up with a differential equation for wave packets?

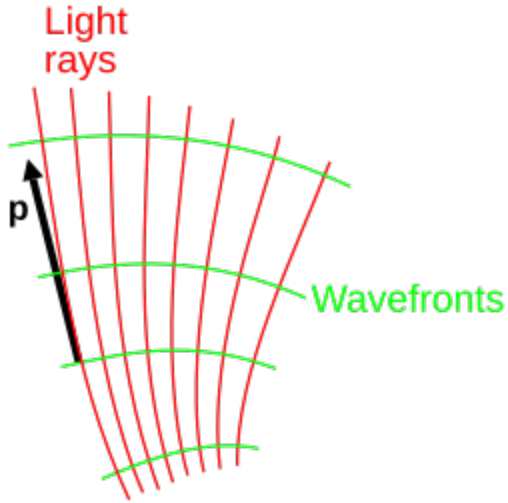
Energy and wave packets

Classical energy

$$E = K + U = \frac{1}{2}mv^2 + U = \frac{p^2}{2m} + U$$

Let's simplify a free particle wave packet as a plane wave, with wave number $k = 2\pi/\lambda$ and angular frequency $\omega = 2\pi f$:

$$\psi(x,t) = Ae^{i(kx-\omega t)}$$



Schrodinger's inspiration: Hamilton's analogy between light as rays/wavefronts and energy as equipotential curves and path of a moving particle

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Apply quantum relationships

Planck/Einstein: $E = hf = \hbar\omega \Rightarrow$

$$\frac{\partial\psi\,(x,t)}{\partial t} = \frac{\partial}{\partial t} A e^{i(kx-\omega t)} = -i\omega\psi(x,t) \Rightarrow E\psi = \; i\hbar\frac{\partial\psi}{\partial t}$$

De Broglie: $p = \frac{h}{\lambda} = \hbar k$

$$\frac{\partial^2\psi}{\partial x^2} = -k^2\psi(x,t) \Rightarrow \; \frac{p^2}{2m}\psi = \frac{\hbar^2k^2}{2m}\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}$$

Multiply the energy equation $E = \frac{p^2}{2m} + U$ by ψ :

$$E\psi = \frac{p^2}{2m}\psi + U\psi$$

Fill in the derivates:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + U\psi$$

Schrödinger's Equation

Erwin Schrödinger
1887-1961



- Describes a particle's interaction with its environment for nonrelativistic quantum mechanics

- Describes not the trajectory, but the ***wave function*** $\psi(\vec{r}, t)$:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}\right) + U\psi$$

- Like Newton's second law, it cannot be derived but is a postulate, to be verified by experimentation

Solving Schrödinger's Equation

- Consider the 1D variant: $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + U\psi$

- Separation of Variables: $\psi(x,t) = \varphi(x) f(t)$

$$i\hbar\, \varphi(x) \frac{df}{dt} = -\frac{\hbar^2}{2m}\frac{d^2\varphi}{dx^2} f(t) + U\varphi(x) f(t)$$

$$i\hbar \frac{1}{f(t)}\frac{df}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\varphi(x)}\frac{d^2\varphi}{dx^2} + U$$

This can only be true if both sides are constant:

$$i\hbar \frac{1}{f(t)}\frac{df}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\varphi(x)}\frac{d^2\varphi}{dx^2} + U = E$$

Solving Schrödinger's Equation

Time component

$$i\hbar \frac{1}{f(t)} \frac{df}{dt} = E$$

$$\frac{df}{dt} = -i \frac{E}{\hbar} f(t)$$

$$f(t) = e^{-\frac{iEt}{\hbar}}$$

Spatial component

$$-\frac{\hbar^2}{2m} \frac{1}{\varphi(x)} \frac{d^2\varphi}{dx^2} + U = E$$

$$-\frac{\hbar^2}{2m} \frac{1}{\varphi(x)} \frac{d^2\varphi}{dx^2} + U = E$$

1D, time-independent Schrödinger Equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2\varphi}{\partial x^2} + U(x)\varphi(x) = E(x)\varphi(x)$$

$U(x)$ → The potential energy of a particle at postion x
$E(x)$ → The total energy (kinetic + potential) of the system, which is constant
$\varphi(x)$ → The wave function that describes the state of a particle with energy E

Schrödinger's Equation – Notation

- From 1D : $-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi(x) = E(x)\psi(x)$
- To 3D: $-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z) + U(x,y,z)\psi(x,y,z) = E(x,y,z)\psi(x,y,z)$

  with $\nabla^2\psi(x,y,z) \equiv \left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right)$

- Hamiltonian operator: $\hat{H} \coloneqq -\frac{\hbar^2}{2m}\nabla^2 + U(x,y,z)$

$$\hat{H}\psi = E\psi$$

The wave function is an eigenfunction of the Hamiltonian operator with corresponding eigenvalue E

- Time-dependent SE: $i\hbar\frac{d\psi}{dt} = \hat{H}\psi$



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Probabilities and Normalization

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi(x) = E(x)\psi(x)$$

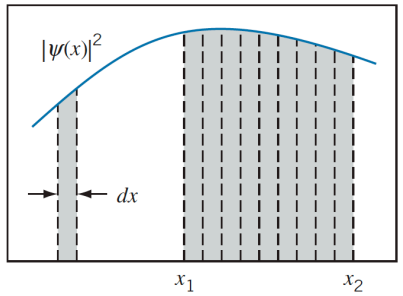
- Wave function $\psi(x,t)$ is *not* the amplitude of the wave packet (ψ is complex and does not describe an actual wave)

- Max Born (1929): $\psi\psi^* = |\psi(x)|^2$ is to be regarded as the **probability density function** of the location of the particle

- The actual probability of finding the particle in a region dx is given by $P(x)dx = |\psi(x)|^2 dx$.

- Normalization: probability of finding the particle anywhere is 1:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$



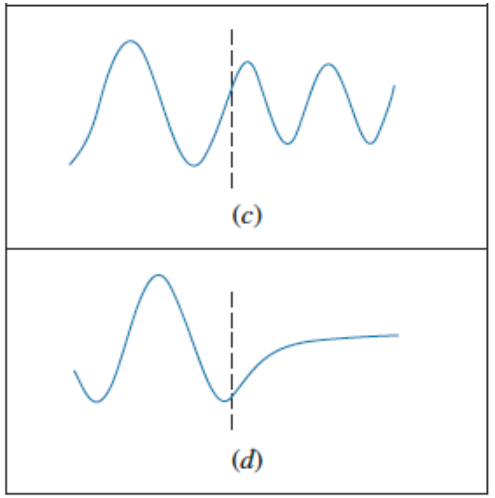
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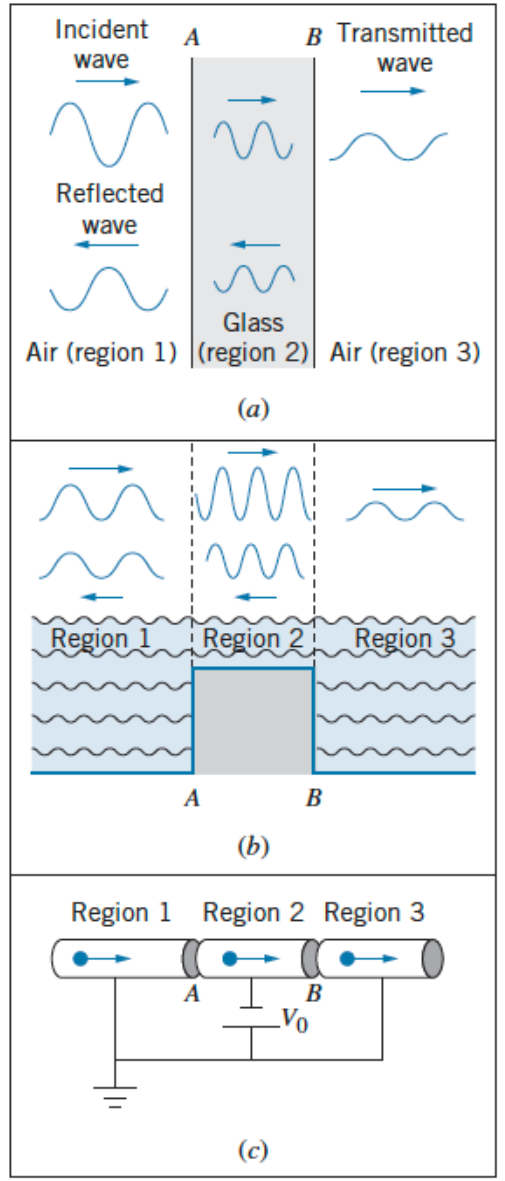
General aspects of waves

When solving Schrodinger's equation, some general aspects of waves apply:

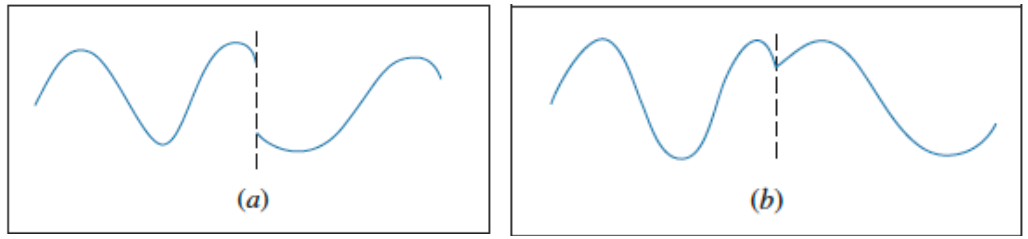
Refraction: part of a wave is reflected at boundaries, part penetrates.

Frequency depends on medium



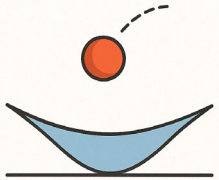


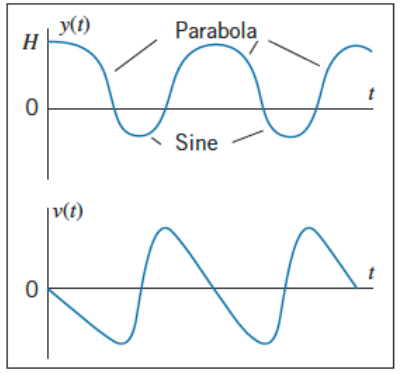
Genral aspects of waves



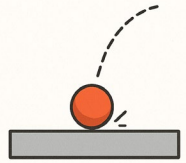
Continuity at the boundaries:

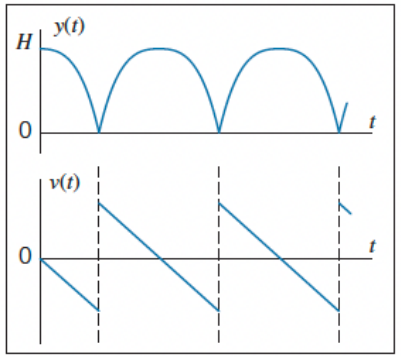
- The wave itself is continuous
- The *slope* (derivative) is continuous (except when the boundary height is infinite)





Position and speed of bal bouncing on rubber sheet (gradual change in potential energy)





Position and speed of bal bouncing on steel surface (step change in potential energy)

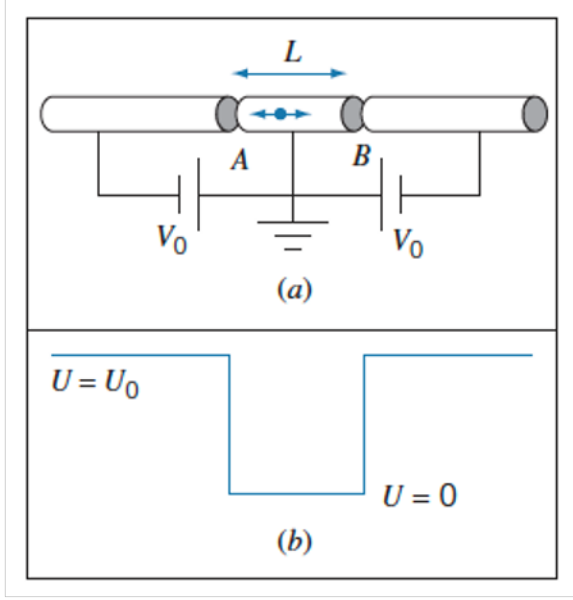
Schrödinger's Equation – Solving Strategy

1. Write down the equation
   - Start from Schrödinger's equation with the given potential $U(x)$
   - For discontinuous $U(x)$, write separate equations for each region
2. Find general solutions
   - Solve the differential equation for $\psi(x)$ using suitable math methods
3. Apply boundary conditions
   - Remove unphysical solutions and determine constants
4. Match at boundaries
   - For discontinuous $U(x)$, ensure $\psi(x)$ (and usually $\frac{d\psi}{dx}$) are continuous at region boundaries
5. Normalize the wave function
   - Require $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$ so total probability is 100%

- Short break

Schrödinger's Equation – Applications

1. Infinite potential energy well ("particle in a box")
2. Finite potential energy well ("particle at a boundary")
3. Free particle

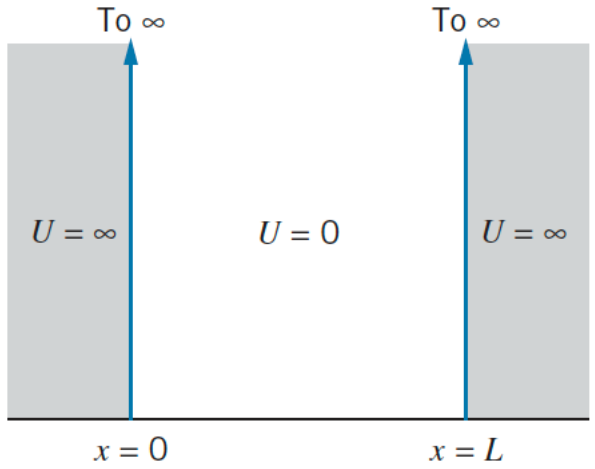


Potential well: an electron trapped between two negative potentials

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Infinite Potential Energy Well

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi(x) = E(x)\psi(x)$$



1) $x < 0, x > L$: $U(x) = \infty$ → Only possible if $\psi(x) = 0$

2) $0 \leq x \leq L$: $U(x) = 0$ →

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = E(x)\psi(x) \Rightarrow \frac{\partial^2\psi}{\partial x^2} = \frac{2mE}{\hbar^2}\psi(x) = -k^2\psi(x)$$

$$\Rightarrow k^2 = \frac{2mE}{\hbar^2}$$

$$U(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty, & x < 0, x > L \end{cases}$$

General solution: $\psi(x) = A\sin(kx) + B\cos(kx)$ ('standing waves')

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Infinite Potential Energy Well

General solution in the well: $\psi(x) = A\sin(kx) + B\cos(kx)$

Boundary conditions: $\psi(0) = 0, \quad \psi(L) = 0$

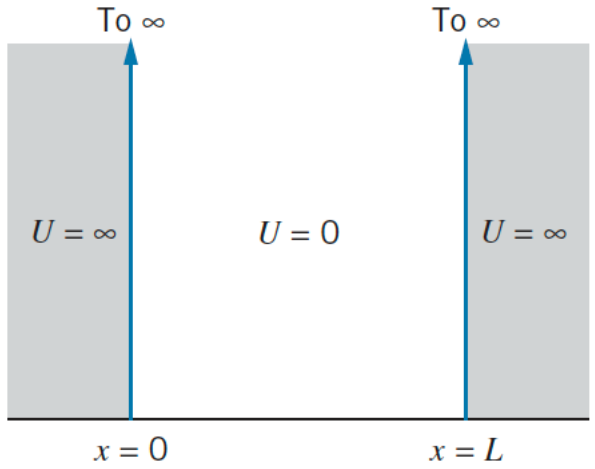
$\psi(0) = 0 \rightarrow \quad \psi(0) = A\sin(k \cdot 0) + B\cos(k \cdot 0) = 0 \Rightarrow \boldsymbol{B = 0}$

$\psi(L) = 0 \rightarrow \quad \psi(L) = A\sin(k \cdot L) = 0 \Rightarrow \boldsymbol{k = n\pi/L}\ (\boldsymbol{n = 1, 2, 3, \ldots})$

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

Using $k^2 = \frac{2mE}{\hbar^2} \rightarrow$ $$E_n = \frac{k^2\hbar^2}{2m} = \frac{n^2h^2}{8mL^2}$$

Only discrete energy levels allowed!



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

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Infinite Potential Energy Well

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right) \qquad E_n = \frac{k^2\hbar^2}{2m} = \frac{n^2h^2}{8mL^2}$$

Find A using normalization constraint:

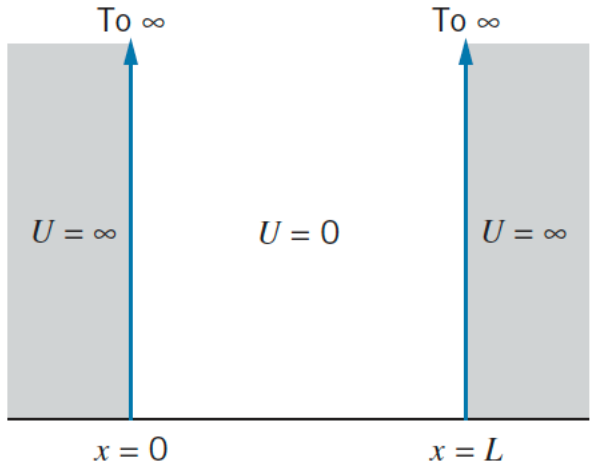
$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = \int_{-\infty}^{\infty} \psi_n \psi_n^* dx = 1$$

$$\int_0^L A^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

By using $\sin^2\alpha = \frac{1-\cos 2\alpha}{2}$:

$$A^2 \int_0^L \frac{1}{2} - \frac{1}{2}\cos\left(\frac{2n\pi x}{L}\right) dx = A^2 \left[\frac{1}{2}x - \frac{1}{4}\frac{L}{n\pi}\sin\left(\frac{2n\pi x}{L}\right)\right]_0^L = 1$$

$$A^2 \left[\frac{1}{2}L - \frac{1}{4}\frac{L}{n\pi}\sin\left(\frac{2n\pi L}{L}\right) - 0\right] = A^2\frac{1}{2}L = 1 \Rightarrow A = \sqrt{\frac{2}{L}}$$



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

$$\boldsymbol{\psi_n(x) = \sqrt{\frac{2}{L}}\, sin\left(\frac{n\pi x}{L}\right)}$$

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Infinite Potential Energy Well

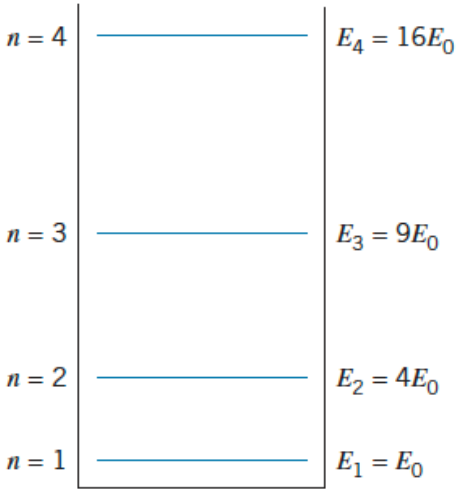


FIGURE 5.10 The first four energy levels in a one-dimensional infinite potential energy well.

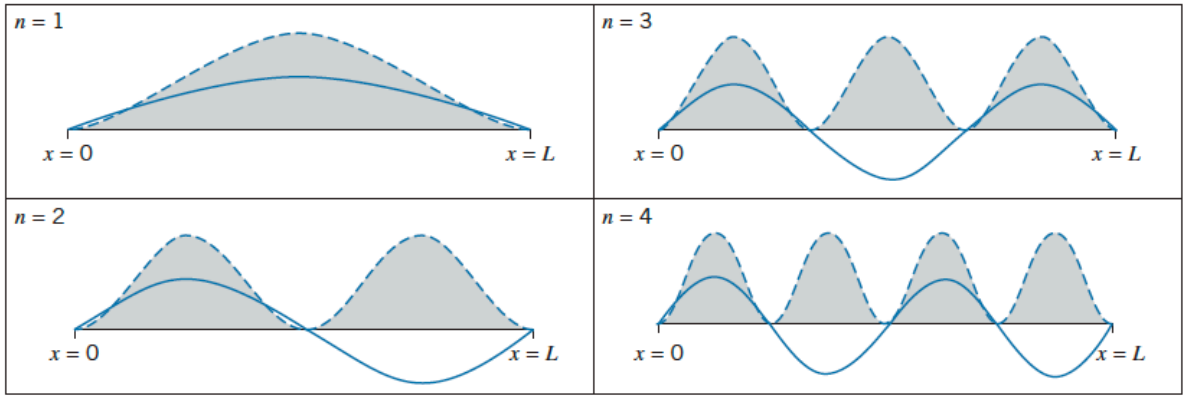
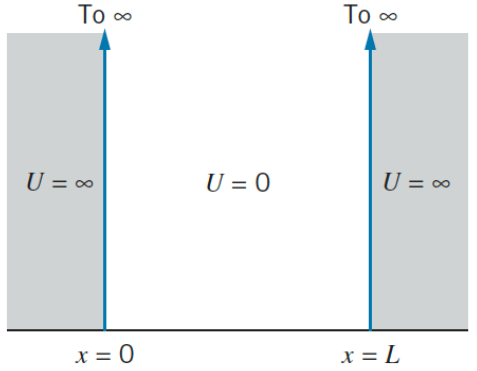


FIGURE 5.11 The wave functions (solid lines) and probability densities (shaded regions) of the first four states in the one-dimensional infinite potential energy well.



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

$$E_n = \frac{k^2\hbar^2}{2m} = \frac{n^2h^2}{8mL^2} = n^2E_0$$

with $E_0 = \frac{\hbar^2\pi^2}{2mL} = \frac{h^2}{8mL^2}$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

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Schrödinger's Equation – Applications

1. Infinite potential energy well ("particle in a box")

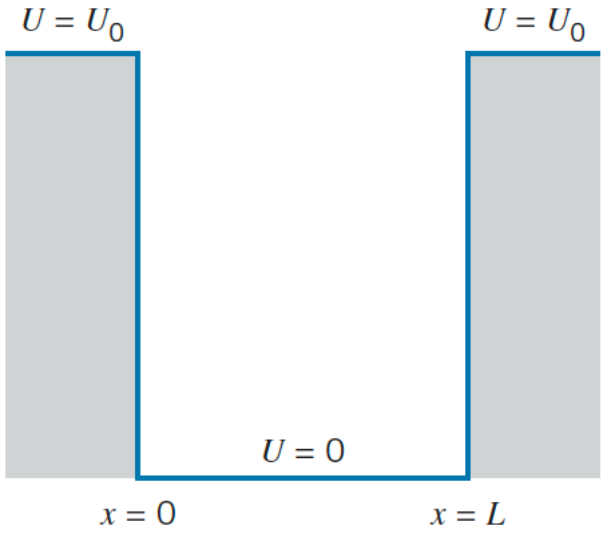
2. **Finite potential energy well**

3. Free particle

Finite Potential Energy Well

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi(x) = E(x)\psi(x)$$

Assume $E < U_0$ (particle is confined)



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ U_0, & x < 0, x > L \end{cases}$$

1) For $0 \le x \le L$, we have the same solution as the infinite well:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

2) For $x < 0$:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = (E - U_0)\psi(x)$$

$$\frac{\partial^2\psi}{\partial x^2} = k'^2\psi(x) \quad \text{with} \quad k' = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$$

$$\psi(x) = Ce^{k'x} + De^{-k'x}$$

$\psi(x)$ should not become infinite for $x \to -\infty$:

$$\psi(x) = Ce^{k'x}$$

3) For $x > L$:

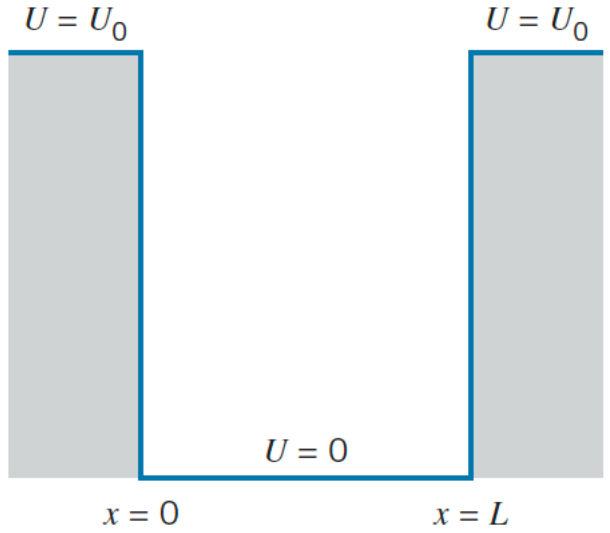
$$\psi(x) = Fe^{k'x} + Ge^{-k'x}$$

$\psi(x)$ should not become infinite for $x \to \infty$:

$$\psi(x) = Ge^{-k'x}$$

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Finite Potential Energy Well



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ U_0, & x < 0, x > L \end{cases}$$

General solution:

1) For $0 \le x \le L$: $\psi(x) = A\sin(kx) + B\cos(kx)$

   with $k = \sqrt{\frac{2mE}{\hbar^2}}$

2) For $x < 0$: $\psi(x) = Ce^{k'x}$
3) For $x > L$: $\psi(x) = Ge^{-k'x}$

   with $k' = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$

Boundary conditions:

1) $\psi(x = 0) = A\sin(k \cdot 0) + B\cos(k \cdot 0) = Ce^{k' \cdot 0} \Rightarrow B = C$
2) $\psi(x = L) = A\sin(k \cdot L) + B\cos(k \cdot L) = Ge^{k' \cdot L}$
3) $\frac{d\psi}{dx}(x = 0) = Ak\cos(k \cdot 0) - Bk\sin(k \cdot 0) = Ck'e^{k' \cdot 0} \Rightarrow Ak = Ck'$
4) $\frac{d\psi}{dx}(x = L) = Ak\cos(k \cdot L) - Bk\sin(k \cdot L) = -Gk'e^{k' \cdot L}$

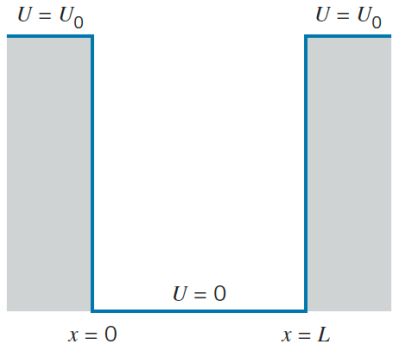
Normalization condition:

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = \int_{-\infty}^{\infty} \psi_n \psi_n^* dx = 1$$

5 equations, 5 unknowns (A, B, C, G)...

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Finite Potential Energy Well



$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ U_0, & x < 0, x > L \end{cases}$$

Numerically solved for $U_0 = 400$ eV and $L = 0.100$ nm

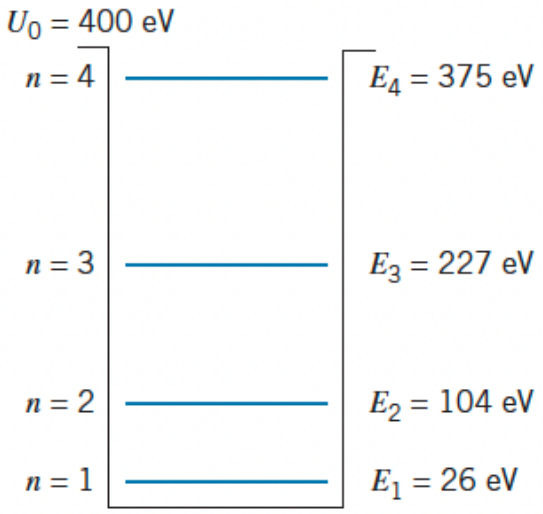


FIGURE 5.13 The energy levels in a potential energy well of depth 400 eV. There are only four energy states in this well.

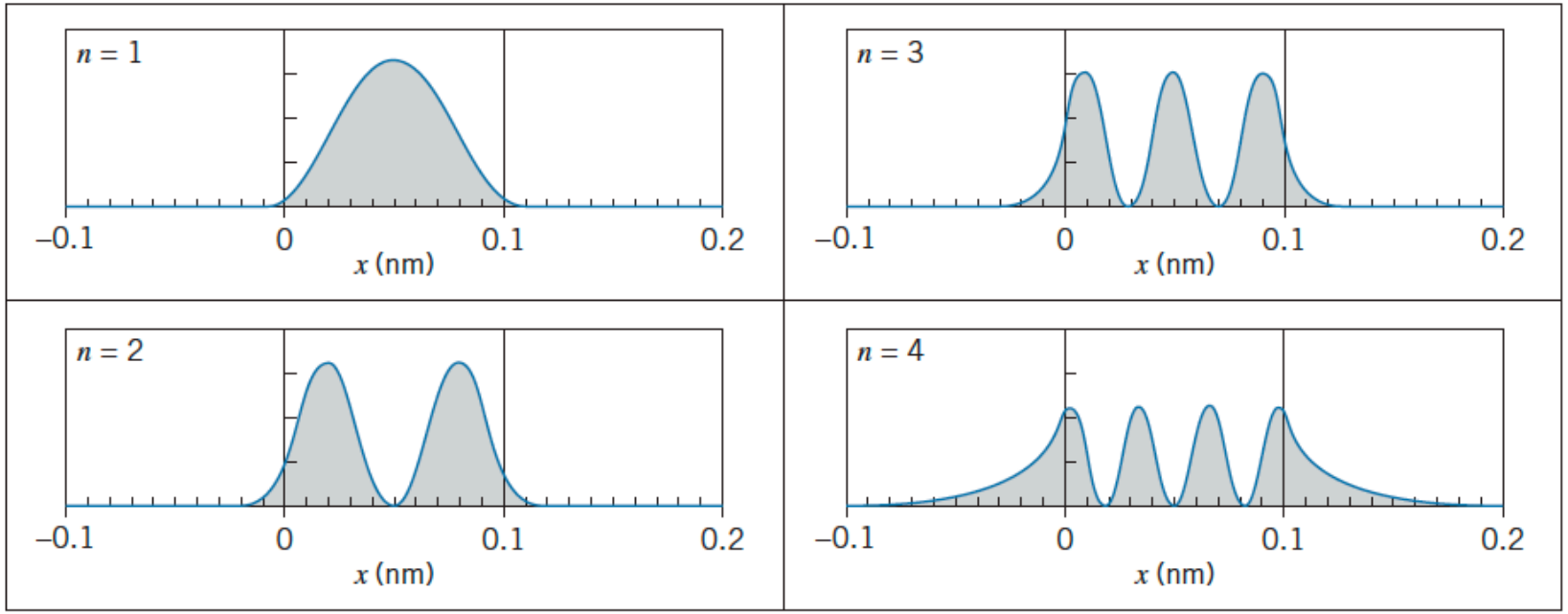


FIGURE 5.14 The probability densities of the four states in the one-dimensional potential energy well of width 0.100 nm and depth 400 eV.

TU/e

Schrödinger's Equation – Applications

1. Infinite potential energy well ("particle in a box")

2. Finite potential energy well

3. **Free particle**

Free Particle

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi(x) = E(x)\psi(x)$$

No force → $U(x)$ is constant, so we can set this to 0 for convenience

$$\frac{\partial^2\psi}{\partial x^2} = \frac{2mE}{\hbar^2}\,\psi(x) = -k^2\psi(x)$$

with $k^2 = \frac{2mE}{\hbar^2}$ and $E > 0$

Solution: $\psi(x) = A\sin(kx) + B\cos(kx)$

We can rewrite this with imaginary exponentials:

$$\psi(x) = A\left(\frac{e^{ikx} - e^{-ikx}}{2i}\right) + B\left(\frac{e^{ikx} + e^{-ikx}}{2}\right) = A'e^{ikx} + B'e^{-ikx}$$

Free Particle

$$\psi(x) = A\left(\frac{e^{ikx} - e^{-ikx}}{2i}\right) + B\left(\frac{e^{ikx} + e^{-ikx}}{2}\right) = A'e^{ikx} + B'e^{-ikx}$$

Adding the time-dependent wave function:

$$\psi(x) = \left(A'e^{ikx} + B'e^{-ikx}\right)e^{-i\omega t} = A'e^{i(kx-\omega t)} + B'e^{-i(kx+\omega t)}$$

For a beam of particles travelling to the right $(+x)$ we use $B' = 0$:

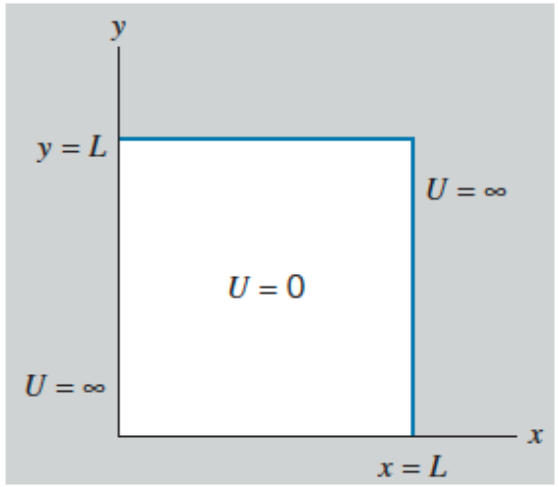
$$\psi(x) = A'e^{i(kx-\omega t)}$$

Normalization:

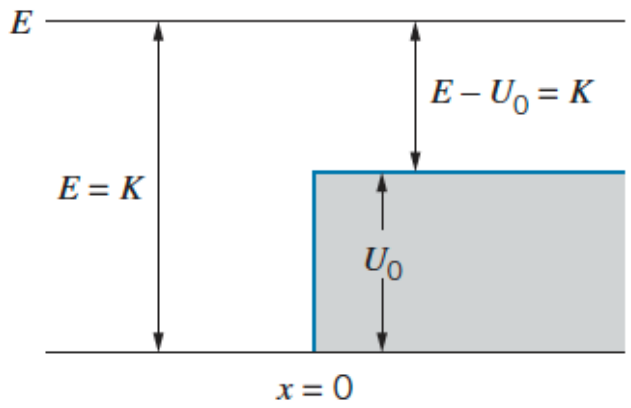
$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} \psi\psi^* dx = |A'|^2 \int_{-\infty}^{\infty} dx = \infty$$

Plane wave cannot be normalized! Corresponds to particle being anywhere

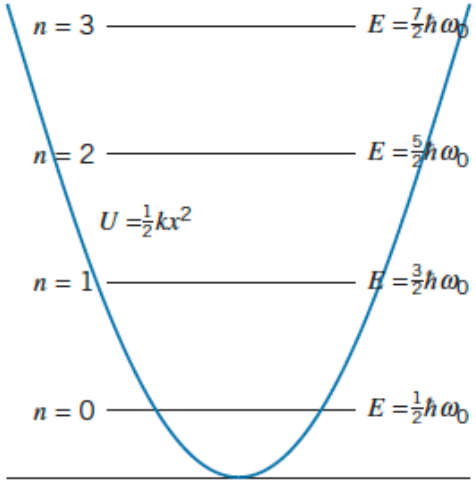
More Potentials…



2D-Infinite Well



Potential Energy Step

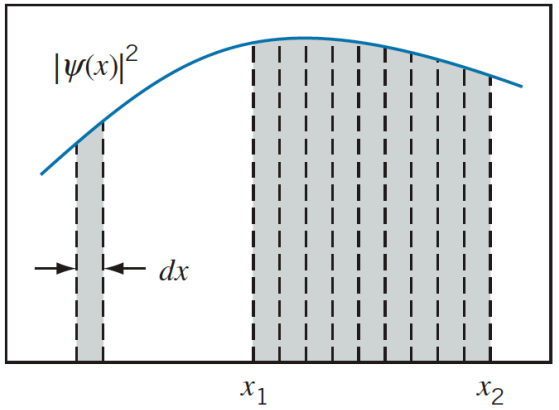


Harmonic Oscillator

- Read and reproduce derivations in Krane
- Practice with Guided-Self Study exercises

Expectation value

- Probability of finding a particle within a region:



$$P(x)dx = |\psi(x)|^2 dx$$

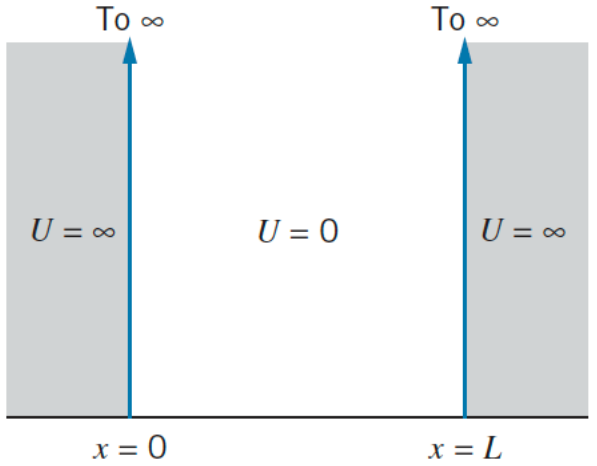
Best we can say is that particle is in region $[x_1, x_2]$ with probability $\int_{x_1}^{x_2} P(x)dx$ ⇒ No two measurements of position x will be the same. Need to revert to weighed value:

- **Expectation value** of any function:

$$\langle f(x) \rangle = \int_{-\infty}^{+\infty} P(x) f(x)\, dx = \int_{-\infty}^{+\infty} |\psi(x)|^2 f(x)\, dx$$

TU/e

Probabilities – Exercise



Reconsider the 1D infinite potential well:

$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

- Show that $\langle x \rangle = \frac{L}{2}$

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

$$\int udv = uv - \int vdu$$

- Calculate :
  - a) the probability of finding the particle between $x = 0$ and $x = \frac{L}{3}$
  - b) The probability of finding the particle between $x = \frac{L}{3}$ and $x = \frac{2L}{3}$
  - c) The probability of finding the particle between $x = \frac{2L}{3}$ and $x = L$

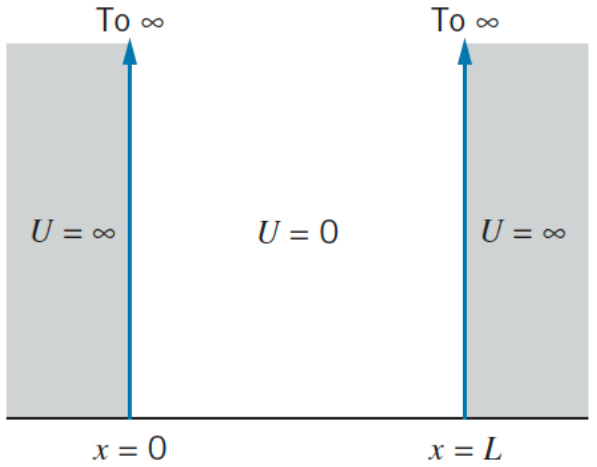
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

For all probabilities, use $n = 1$

TU/e

Probabilities – Exercise

$$\sin^2\alpha = \frac{1-\cos 2\alpha}{2}$$



$$\langle x\rangle = \int_{-\infty}^{+\infty} P(x)x\,dx = \int_{-\infty}^{+\infty} |\psi(x)|^2 x\,dx = \int_0^L \frac{2}{L} x \sin^2\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L}\int_0^L x\left[\frac{1}{2} - \frac{1}{2}\cos\left(\frac{2n\pi x}{L}\right)\right] dx = \frac{1}{L}\left[\int_0^L x\,dx + \int_0^L x\cos\left(\frac{2n\pi x}{L}\right) dx\right]$$

$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

$$\int_0^L x\,dx = \left[\frac{x^2}{2}\right]_0^L = \frac{L^2}{2}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

$$\int_0^L x\cos\left(\frac{2n\pi x}{L}\right) dx = \left[x\left(\frac{L\cdot\sin\left(\frac{2n\pi x}{L}\right)}{2n\pi}\right)\right]_0^L - \int_0^L \frac{L\cdot\sin\left(\frac{2n\pi x}{L}\right)}{2n\pi} dx = 0$$

Integration by parts:

$$\int udv = uv - \int vdu$$

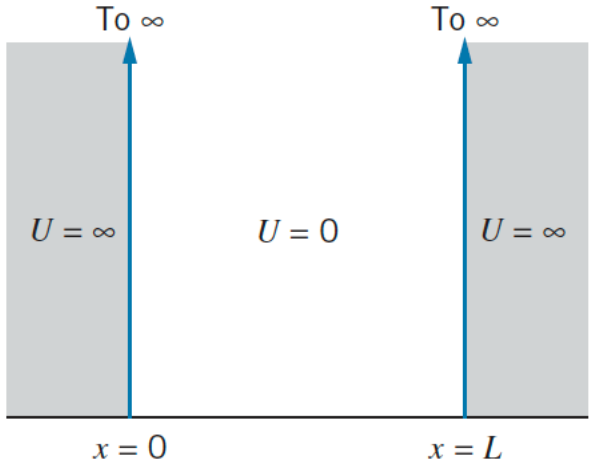
$$\langle x\rangle = \frac{1}{L}\left[\frac{L^2}{2} + 0\right] = \frac{L}{2}$$

TU/e

Probabilities – Exercise

$$\sin^2\alpha = \frac{1-\cos 2\alpha}{2}$$

Using $\frac{\pi x}{L} = u,\ du = \frac{\pi dx}{L} \Rightarrow dx = \frac{L}{\pi}du$



$$P\left(0{:}\frac{L}{3}\right) = \int_0^{\frac{L}{3}} |\psi_1(x)|^2 dx = \int_0^{\frac{L}{3}} \frac{2}{L}\sin^2\left(\frac{\pi x}{L}\right) dx = \frac{2}{\pi}\int_0^{\frac{\pi}{3}} \sin^2(u)\, du = \frac{2}{\pi}\left[\frac{u}{2} - \frac{\sin(2u)}{4}\right]_0^{\frac{\pi}{3}} = 0.1955$$

$$U(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & x < 0, x > L \end{cases}$$

$$P\left(\frac{L}{3}{:}\frac{2L}{3}\right) = \frac{2}{\pi}\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \sin^2(u)\, du = \frac{2}{\pi}\left[\frac{u}{2} - \frac{\sin(2u)}{4}\right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}} = 0.6090$$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

$$P\left(\frac{2L}{3}{:}L\right) = \frac{2}{\pi}\int_{\frac{2\pi}{3}}^{\pi} \sin^2(u)\, du = \frac{2}{\pi}\left[\frac{u}{2} - \frac{\sin(2u)}{4}\right]_{\frac{2\pi}{3}}^{\pi} = 0.1955$$

TU/e

Barriers

- Classical solution: particle with $E < U_0$ cannot exist in $x > L$
- Exponentially dropping real part in barrier, transmitted wave afterwards
- ‘tunneling’ through barrier
- Particle cannot be observed (‘exist’) inside barrier ($\Delta E\ \Delta t > \hbar$)

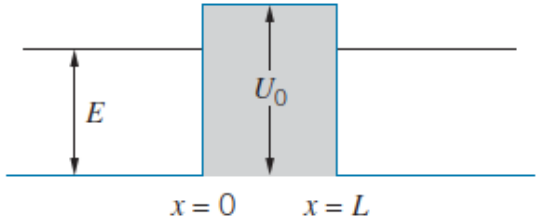


FIGURE 5.26 A barrier of height U_0 and width L.

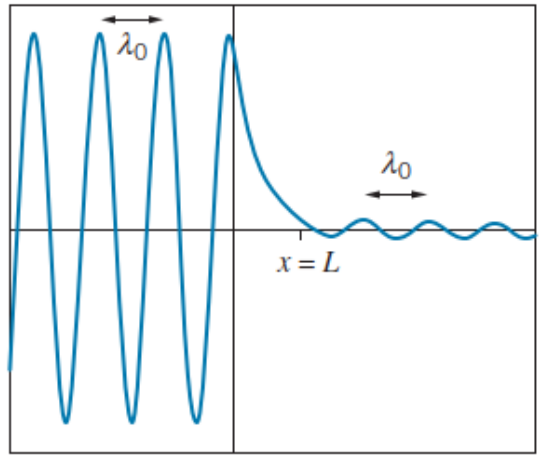
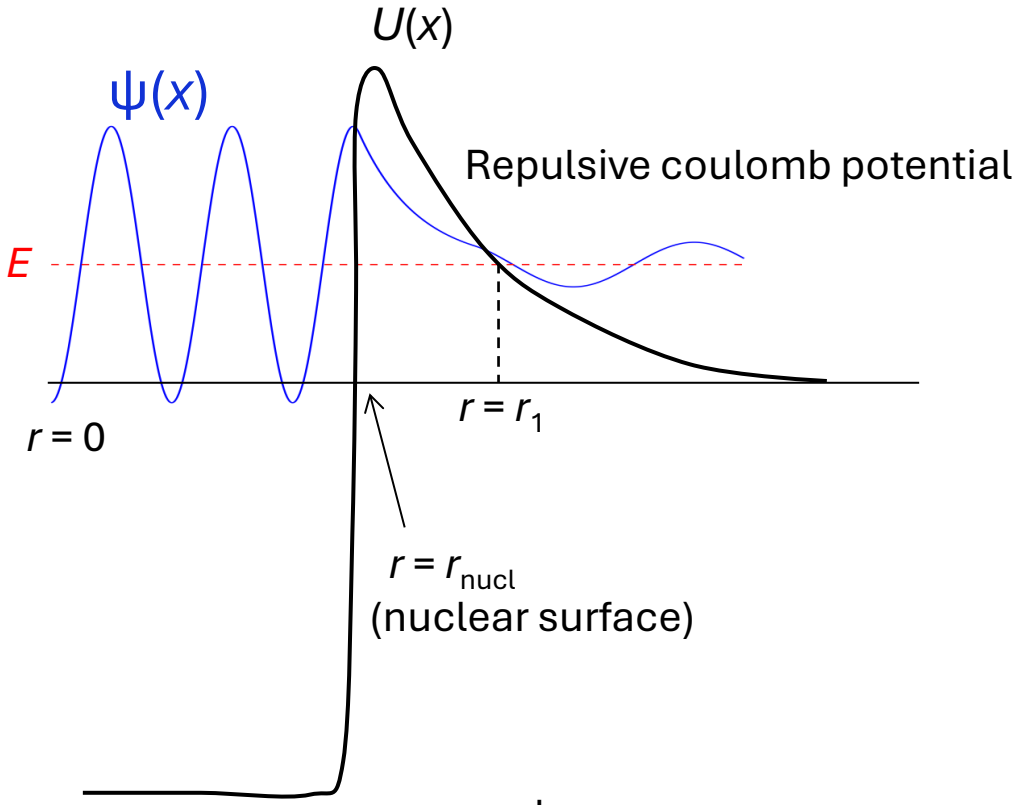


FIGURE 5.27 The real part of the wave function of a particle of energy $E < U_0$ encountering a barrier (the particle is incident from the left in the figure). The wavelength λ_0 is the same on both sides of the barrier, but the amplitude beyond the barrier is much less than the original amplitude.

TU/e

Potential Barriers & Tunneling – α Particle



α decay =
emission of α particle (He nucleus) by a (usually heavy) nucleus

For $E > U(x)$:

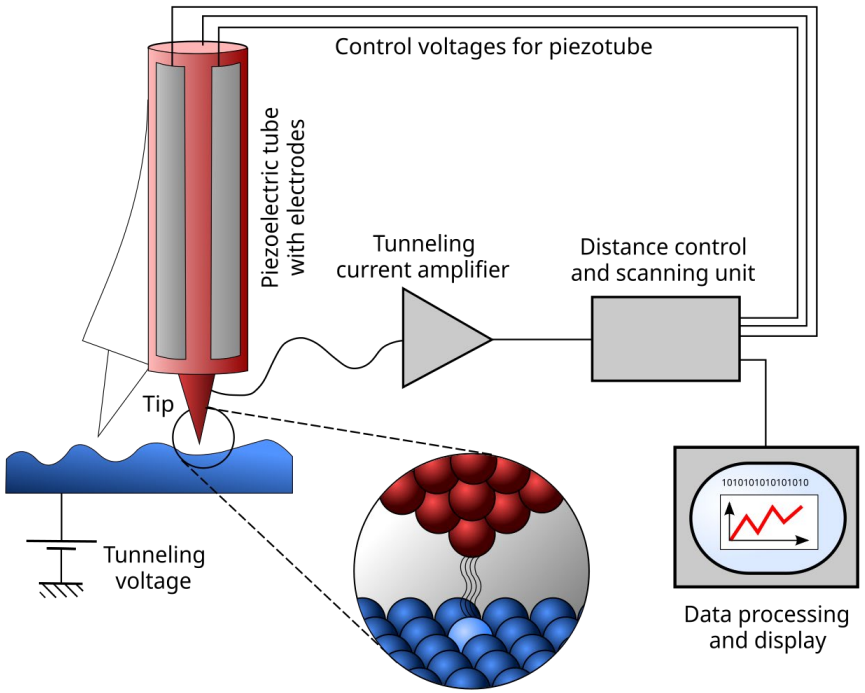
- Wave function has oscillatory form in two regions: $r < r_{nucl}$ and $r > r_1$

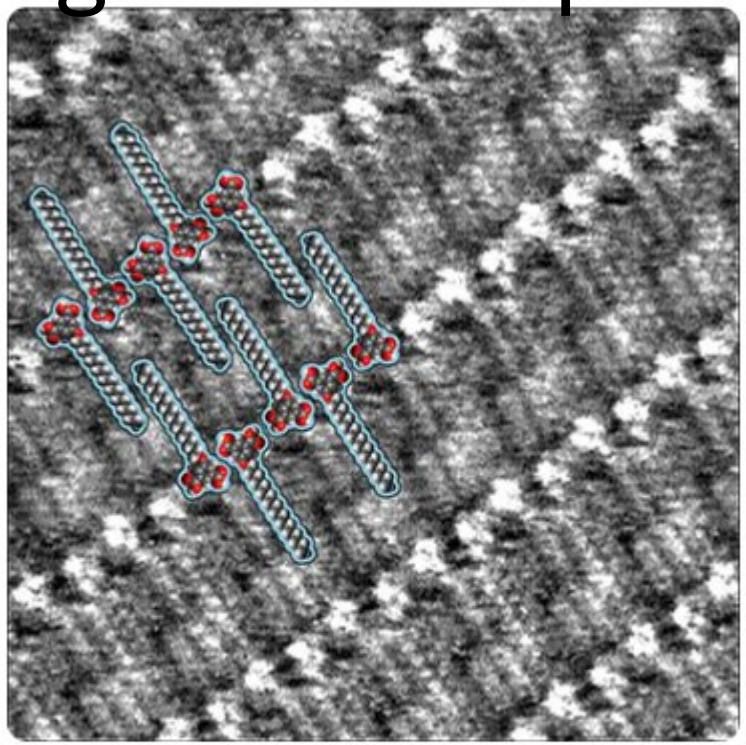
For $E < U(x)$:

- Wave function has exponential decay in region: $r_{nucl} < r < r_1$
- Classical approach: the particle has insufficient energy to cross the barrier
- Quantum mechanics: the particle can *tunnel* through this barrier and escape from the nucleus

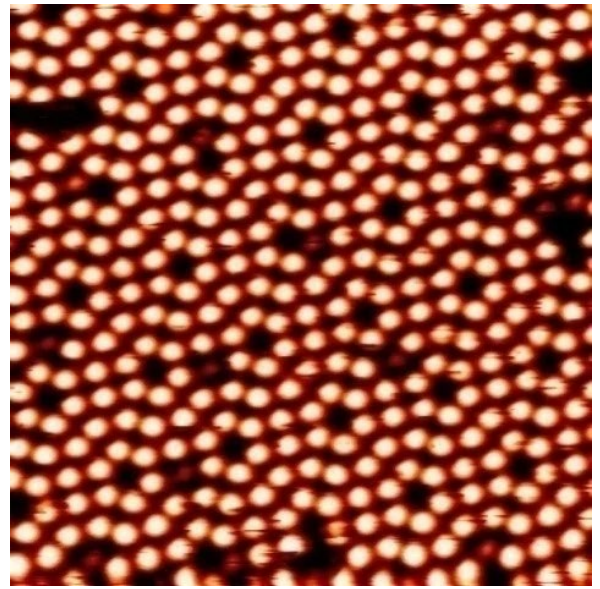
TU/e

Scanning Tunneling Microscope



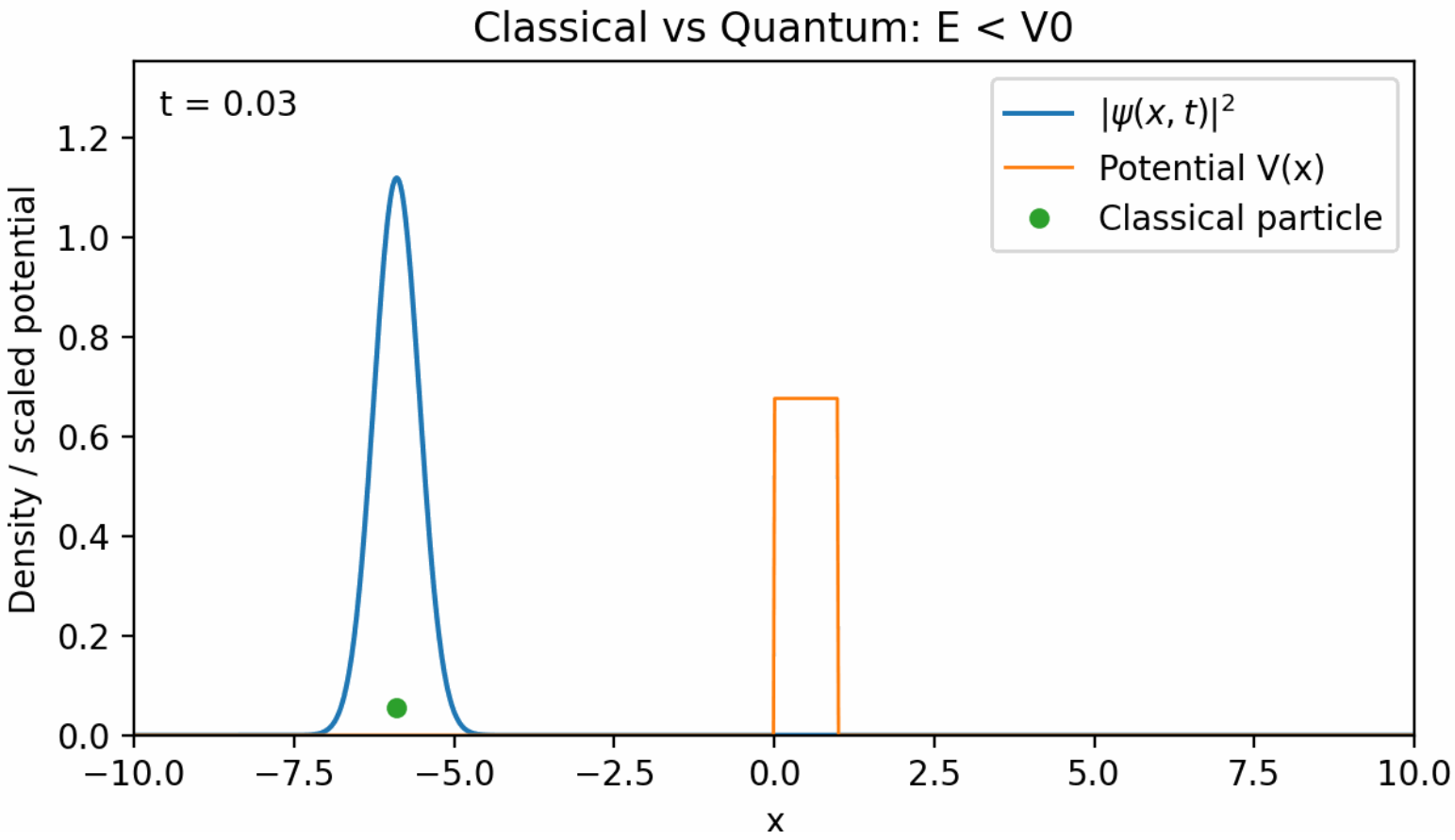


STM image of a self-assembled monolayer of 5-octadecoxyisophthalic acid on highly-oriented pyrolytic graphite.
Molecular models are overlaid to emphasize the molecular ordering. Imaged on a Cypher AFM, 15 nm scan. Sample courtesy of Matt Blunt, University of Nottingham.



Atoms on a silicon surface

Potential Barriers & Tunneling



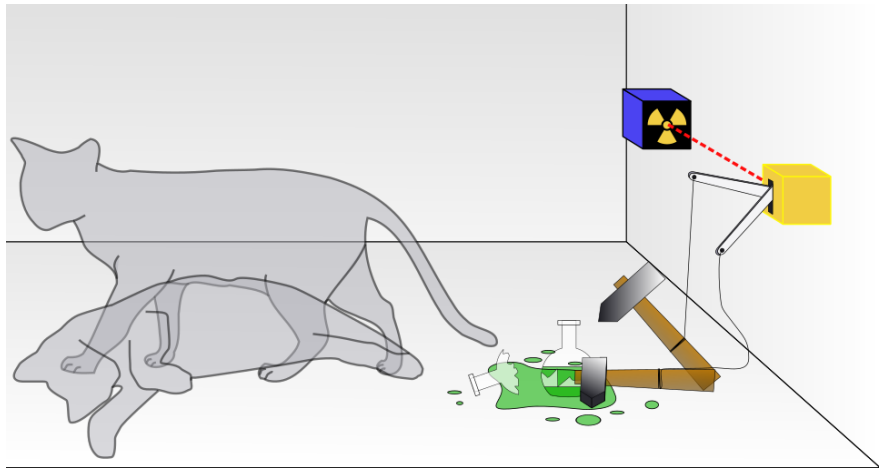
TU/e

Schrodinger’s cat

- Double slit experiment: ‘unobserved’ particles show diffraction, ‘observed’ particles show lines: why??

- Copenhagen interpretation: ‘particle is in all possible states at once, until measurement forces it to choose 1 state (lines): *waveform collapse*’

Schrodinger's thought experiment



- A cat is locked in box. The box contains a radioactive atom. Once that decays, a geiger counter will detect it and cause a chain of events killing the cat.
- Copenhagen: the atom is in both the undecayed and decayed state while unobserved. By extending this reasoning *in absurdo*, the cat is both alive and dead while unobserved.
- Schrodinger used this thought experiment to show our interpretation was incomplete or flawed when extending quantum physics to the real world